

Introduction to Using the Arduino Development System

ref1: <https://docs.arduino.cc/hardware/uno-rev3/>

Student Notes

Table of Contents

0. Setting up the Arduino IDE.....	1
1. Blink; the hardware “Hello World” application.....	1
2. How the system works.....	2
3. LEDs.....	2
4. Adding external LEDs to the Arduino aboard.....	3
4a. Turn each LED ON.....	3
4a1. Straight line code.....	4
4a2. Using an Array.....	4
5. 7 Segment LEDs.....	4
5a. Single 7-Segment LED Display.....	4
5a1. Using Arduino functions – sketch LED7seg_5a1.ino.....	4
5a2. Using direct hardware control registers (PORTD).....	5
5b. Multiple 7-Segment LED Displays.....	5
6. I2C Control.....	6
6a. I2C Master Controller.....	6
6b. I2C Slave Device.....	6

0. Setting up the Arduino IDE

Go to the Arduino website at

<https://www.arduino.cc/> > For Makers > Go to Documentation > Software > Arduino IDE 2

There is a walk-through for installing the Arduino IDE and hardware plus lots of other valuable information about the Arduino product line, project ideas, and additional help.

NOTE: Some “Nano” boards, like those from DFRobots, use the “Arduino Duemilanove or Diecimila” profile.

1. Blink; the hardware “Hello World” application.

This program can be found in

File > Examples > 01.Basics > Blink

Click Blink to load the program into the IDE. Connect your Arduino board to the computer using the USB cable. In the pull-down “Select Board” in the menu bar, select the board connected if the system has not already set this for you.

This program is stored in the installation folders and these should not be changed. So the first thing to do, if you want change this program is to save into your own folder. Use File > Save As... and select the path for your folder. In the folder, unclick the read-only file property of the Blink.ino file and load the file from your folder into the IDE for editing. I use “C:\Users\<your name>\Documents\Arduino” for my files.

On line 34, change the delay(1000) to delay(100). Click the Upload arrow and the LED should now Blink quickly each second. This shows that you have control of the hardware though your programming.

The Arduino is programmed (normally) using the C/C++ programming language. Learning to program the Arduino will be done as in this example; modifying existing code. Most programming is done this way; by stating with a template or existing project.

2. How the system works

On RESET, the system will perform some initialization and then jump to the `setup()` function in the downloaded code. After exiting the `setup()` function, it then enters a continuous cycle of calling the `loop()` function then servicing its internal operation then calling the `loop()` function, etc. This is why it is important to always return from the `loop()` function.

3. LEDs

Some electronics theory is now needed so that parts (the Arduino or the LEDs) are not damaged.

The first question to be answered is “What does it take to light up an LED?”. The standard red LED requires between 1.8v and 2.2v across it and at least 1ma of current available. The schematic symbol and physical case are shown in Figure 1. The FLAT side of the circular LED case indicates the Cathode or negative terminal.

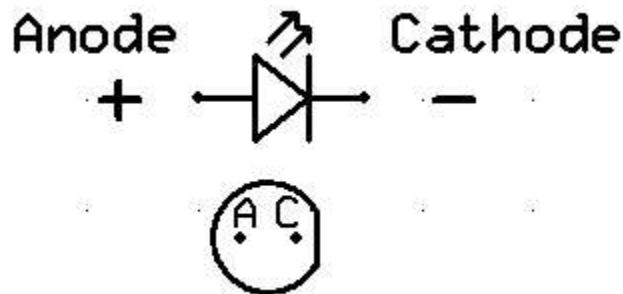


Figure 1

The Anode is connected to the positive voltage source and the Cathode is connected to the negative or ground side. Current (by convention) flows from positive to negative.

At 1ma., it will be very dim, but will light up. Most LEDs can handle 20ma and up to 30ma for a short duration. We will assume that the LED drops 2v and will use 10ma for brightness. Brightness can be controlled by how much current is used, but the voltage drop across the LED will not change.

Looking at the Arduino UNO 3 hardware specifications (see ref1 Tech Specs), the Arduino I/O pins can supply 5V @ 20ma.

The problem then is provide the correct voltage to the LED at a desired current. The solution is to use a resistor in series with the LED and Ohm's Law to calculate the resistor's value.

This is a simple circuit for the Arduino connected to an LED.

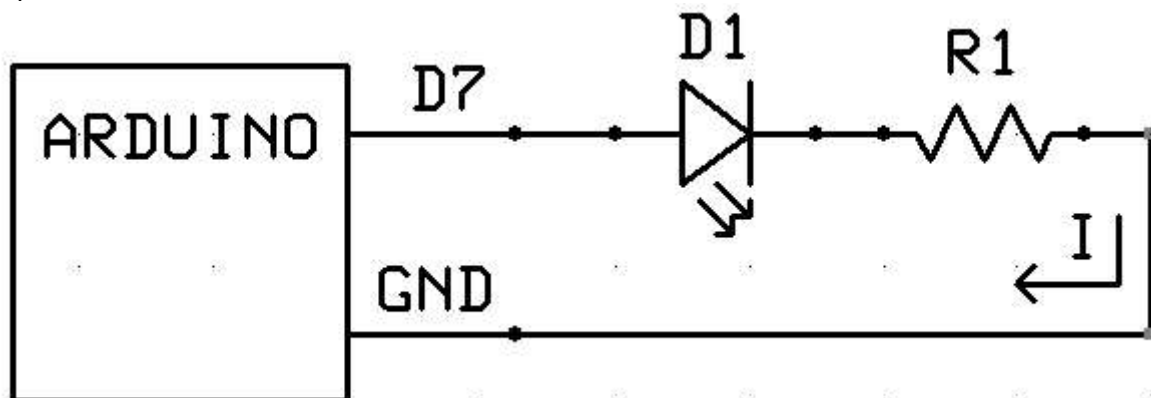


Figure 2

It is a 'series' circuit because all of the components are strung together in a series, one after another. There's another law called Kirchoff's Voltage Law (KVL) that states that "The sum of the voltages (drops and rises) in a series circuit must add up to zero." Another Kirchoff Law states the "Current entering a node equals the current exiting a node." In a series circuit, each connection is a node with one input and one output. Therefore all of the connection have the same current (I) flowing though it and each component has the same current flowing though it. (i.e. the current is the same everywhere in the circuit).

Following the circuit in Figure 2, the voltage starts a +5v (Arduino I/O Pin), then drops 2v across the LED, then drops V_r across the resistor, and ends up at 0v and Ground (GND). Using KVL, $5 - 2 - 3 = 0$, so the resistor must drop 3 volts. Using Ohm's Law that $E = IR$ or Voltage = Current x Resistance, the known values are plugged into the equation. $V_r = I \times R$. V_r and I are know. R is to be solved for. Rearrange the equation to solve for R .

$$V_r / I = R \rightarrow 3v / 10ma = 3v / 0.01A = 300 \text{ ohms.}$$

The nice thing about LEDs is that their brightness is subjective; a little dimmer or a little brighter is hardly noticeable. If a 300 ohm resistor is not available, use a 270 ohm (11ma) or a 330 ohm (9ma) resistor. These are common values for resistors. Other common values are 150, 220, 390, and 470. Any of these will cause the LED to light up.

4. Adding external LEDs to the Arduino aboard.

Apply information from #2 to preform experiments with six LEDs. Connect up to six LEDs to the Arduino board as shown in Figure 3. Different resistor values may be used based on availability or desired brightness.

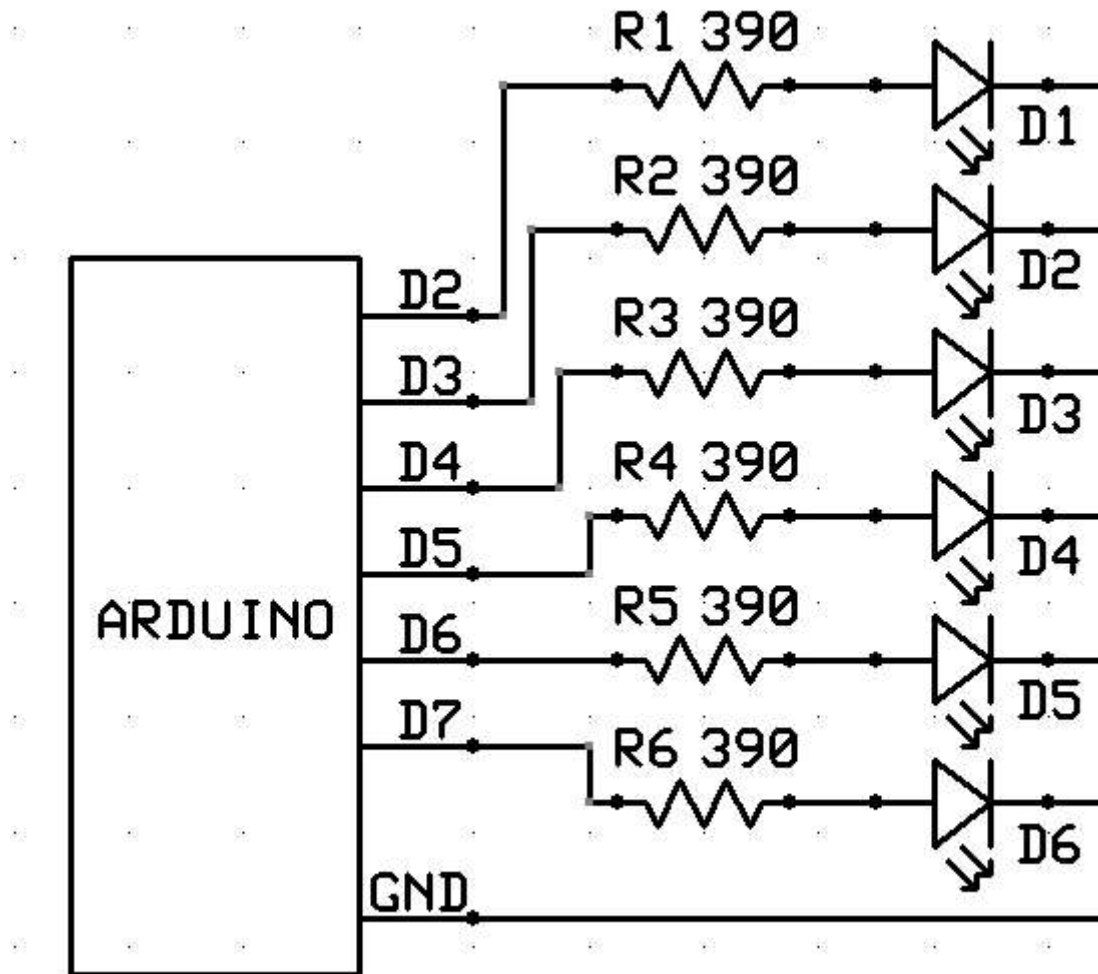


Figure 3

4a. Turn each LED ON; one at a time, then turn each LED OFF, one at a time.

In programming, there are many ways to do the same thing. The follow section will explore this concept.

4a1. Straight line code.

It's usually easiest to start with a straight-line code approach to any new problem. This can provide insight into how the hardware controls an external device and logical issues of control.

Using the Blink sketch as a reference, a few things are needed to control an LED connect to an I/O pin.

Use File > Open > the LED_3a1 file, located in the ITA_Examples folder provided, to load the example code into the IDE. Read though the code and use the Help > Reference to understand the instructions.

4a2. Using an **Array** to control which LED is Blinked.

Use File > Open > the LED_3a2 file.

4a3. Using an Array of **Data Structures** {LED#, State, time} to control output and delay.

Use the Array as a circular pattern generator.

Use the Array as a scrolling patter generator; Chase, Ping-pong, Morse Code.

4a4. Using the random function to randomly pick an LED from the Array to turn ON/OFF or Toggle.

Use an Array to record the **State** of an LED.

4b. Use a **State Machine** to cycle through different display types.

3b1. Use time checks to set the run-time for each state.

3b2. Use a **Table** to control the state sequence.

5. 7 Segment LEDs

5a. Single 7-Segment LED Display

5a1. Using Arduino functions – sketch LED7seg_5a1.ino

A 7-segment LED is just 8 LEDs arranged to display characters. The anode of the LED is attached to it's respective letter pin (a, b, c, etc.) and all LED cathodes are attached to both of the C (common) pins 3 and 8. So, the connections are the same as in Section3, except for eight LED instead of 6. The other difference in the hardware is the use of a transistor to sink the current from all of the LEDs. An Arduino pin can only sink 20ma. If all of the LEDs are ON, then the common line could have as much as 160ma which is beyond the Arduino's capability. Therefore, a NPN transistor is used.

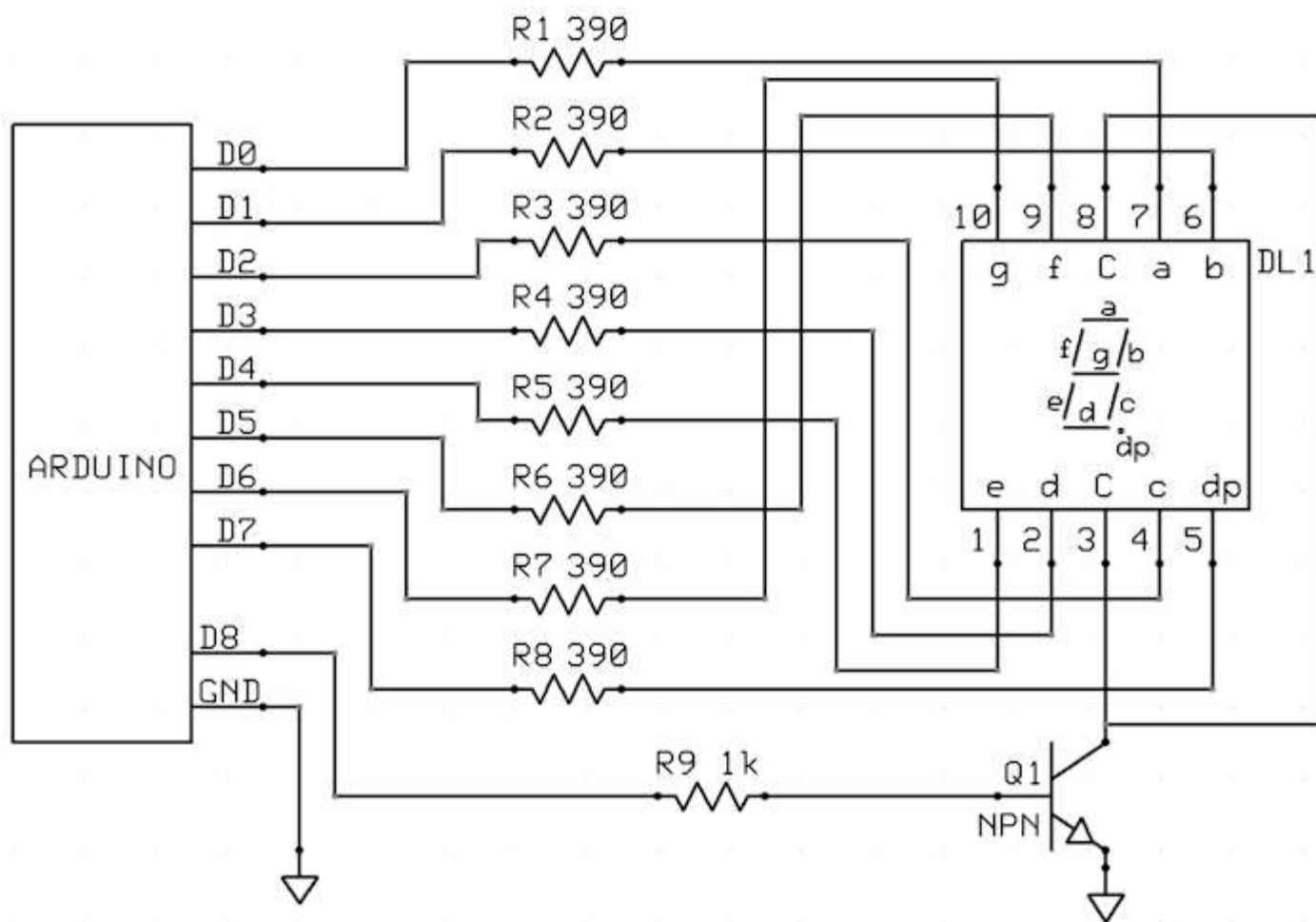


Figure 5

5a1a. Removing Delay() calls.

5a2. Using direct hardware control registers (PORTD)

Open the file LED7seg_4a2 in the IDE and review the code. Upload the sketch to the Arduino and the display should cycle through the counts.

5b. Multiple 7-Segment LED Displays

5b1. Using the Multi Display Demo board. Connect Arduino Digital pins D0:D7 to the J1 Segments header with D0 for seg_a, etc. per the schematic. Connect D8 to J1 pin 9 to control the first 7-segment LED. Connect D9 to J1 pin 10 to control the second digit.

Open the file MULTI7seg_4b1 in the IDE and review the code. Upload the sketch to the Arduino and the displays should cycle through each digit. More displays can be connected in a similar manner until there are no more available digital pins.

5c. Controlling Brightness with Software

5d. More and More LEDs

5d1. How to string LEDs in series.

Two LEDs or more LEDs with higher supply voltage.

5d2. Add SN74LS138N to control eight LEDs with three lines.

Add a fourth line to be able to turn all LEDs off.

5d3. Control twenty LEDs with five lines.

Differential LED drive (Charlie-plexing)

6. I2C Control

The Arduino can act as either a Master Controller or as a Slave Device. This section will start with an I2C controllable temperature sensor.

6a. [I2C Master Controller](#)

6b. I2C Slave Device